

		$m(g\sin\theta) = 5m$, so $\theta = \sin^{-1}\left(\frac{5}{9.81}\right) = 30.6^\circ$
6	C	Let W be the weight on the board When the child is not on the board, $F(2.0) = W(2.5)$ The force on the spring $F = 1.25W$ When the child is on the board, $F'(2.0) = W(2.5) + (40)(9.81)(5.0)$ The force on the spring $F' = 1.25W + 981$ Hence, the addition force on the spring due to the child is 981 N. The additional compression on the spring $= 981/10000$ $= 9.8 \text{ cm}$
7	B	Additional energy stored in spring $= \frac{1}{2} \times 10^4 \times (0.0098^2 - 0.004^2) = 0.05789 \text{ J}$